

Dhayaanidhi Jagadesan Govindharaju

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Education

Masters in Embedded Systems

Technische Universität Chemnitz, Germany *10/2024 – Present*

Key courses: Design of Software for Embedded Systems, Project Lab Embedded Systems, Design of Digital Systems, Hardware/Software-Co-design, Automotive Sensor Systems

Bachelors in Electronics and Communication Engineering

Sri Shakthi Institute Of Engineering and Technology, Coimbatore, India *08/2016 – 11/2020*

Languages

Deutsch: B1, **English**: C1, **Tamil**: C2, **Kannada**: B2

Academic Projects

Battery *SoC Estimation using AI & Physics-Based Methods:*

Python, Scikit-learn, TensorFlow, EIS data

Implemented and benchmarked few ML models (Extra Trees, XGBoost, CNN-1D) on LiFePO4 battery impedance data, achieving 0.458% MAE compared against physics-based BMS approaches for embedded deployment suitability.

Smart Healthcare System:

Embedded C, ADC sensors, ATMEGA microcontroller

Designed an IoT-based patient monitoring system which collects vitals and sends to database which later visualized by doctor and care taker in mobile application with alert feature for real-time health tracking.

Experience

Embedded Linux Developer Intern

AUMOVIO Autonomous Mobility Germany GmbH, Ingolstadt, Germany *08/2026 – Present*

Projects: Modular Web Platform for Embedded Linux Deployment & Diagnostics

- Architected a four tier modular web platform using SOLID principles, improving UI responsiveness on a single core processor by replacing blocking shell scripts with asynchronous job queues
- Developed C/C++ firmware updates for client devices, adding new functionalities to handle network commands that publish device information
- Directed an agentic AI coding workflow to build the web UI, accelerating frontend development while maintaining strict code quality and focusing manual engineering on the core backend.
- Implemented device discovery using DHCP leases, periodic availability monitoring, CSV export functionality, and firmware upgrades synced with Git release tags.
- Set up an eight step SDLC pipeline following IEEE 1012 standards and measured architectural improvements in coupling and complexity using ISO/IEC 25010 metrics

Associate Consultant (Embedded Software Engineer)

Capgemini Technology Services, Bengaluru, India *07/2023 – 10/2024*

Projects: Volkswagen EU7 Engine ECU

- Engineered and integrated AUTOSAR compliant ASW and BSW modules following ASPICE standard for Volkswagen EU7 powertrain engine ECUs.

- Accelerated software validation by generating HEX and A2L files, actively resolving complex build and integration bottlenecks.
- Executed rigorous system validation, calibration and fault analysis by performing Hardware-in-the-Loop (HiL) testing with LabCar simulators.
- Handled customer documents and check lists assuring proper customer delivery.

Firmware Engineer

Sacra Systems Private Limited, Coimbatore, India *01/2022 – 05/2023*

Projects: Smart Hot Beverage Vending Machine

- Built fault-tolerant FreeRTOS firmware for STM32-L microcontrollers, using state machines and object-oriented modules to handle precise motor control, drink preparation, and automated cleaning
- Wrote low-level peripheral drivers (ADC, DAC, PWM, SPI, I2C, UART) and a custom flash data manager to handle sensor control and NAND NVS data.
- Developed multiple object oriented application level modules for drink preparation, self-cleaning, real-time error detection and handling, machine calibration, etc.,
- Integrated AWS IoT Core (MQTT) to stream real-time machine telemetry and set up a secure Over-The-Air (OTA) firmware upgrade pipeline.
- Redesigned the C++ and Qt-based UI application running on a Linux Raspberry Pi to improve the user experience based on customer feedback.
- Directed a 3-person engineering team, managing code reviews and translating technical customer requirements into deployable features.

Junior IoT System Design Engineer

Auxo Technology Labs Private Limited, Chennai India *12/2020 – 01/2022*

Projects: Smart Healthcare Ambient Monitoring Device, Food processing machinery

- Engineered edge computing architecture (ESP32, Raspberry Pi), developing object-oriented C/C++ modules for audio capture, storage, and secure streaming.
- Programmed BLE protocols and Python based Linux scripts to continuously stream device telemetry and handle secure edge-server authentication.
- Deployed encrypted Over-The-Air (OTA) updates utilizing server-side binary packages to ensure seamless remote maintenance.
- Rapidly prototyped firmware for crucial Proof of Concepts (POCs) and Minimum Viable Products (MVPs), driving early-stage product validation.
- Designed secure MQTT based IoT architectures, taking ownership of system connectivity and real-time data streaming.

Technical Skills

- **Programming skills:** C/C++, Python
- **Microcontrollers & Processors:** STM32 (L and H series), ESP32, ATMEGA, Raspberry Pi
- **Operating System & Frameworks:** Embedded Linux (Buildroot), FreeRTOS, AUTOSAR
- **Peripherals, Protocols and Networking:** SPI, I2C, UART, PWM, ADC, DAC, SDIO, CAN, LIN, MQTT, NTP/PTP, DHCP/DNS/TFTP
- **Software Tools:** VSCode, INCA, UDE, STM32CubeIDE, STM32CubeProgrammer, lighttpd, dnsmasq
- **Hardware Tools:** Circuit analysis, J-Link/ST-Link Debug Probes, Oscilloscope, Multi-meter
- **Cloud / Version Control:** Git, AWS IoT, AWS CodeCommit.